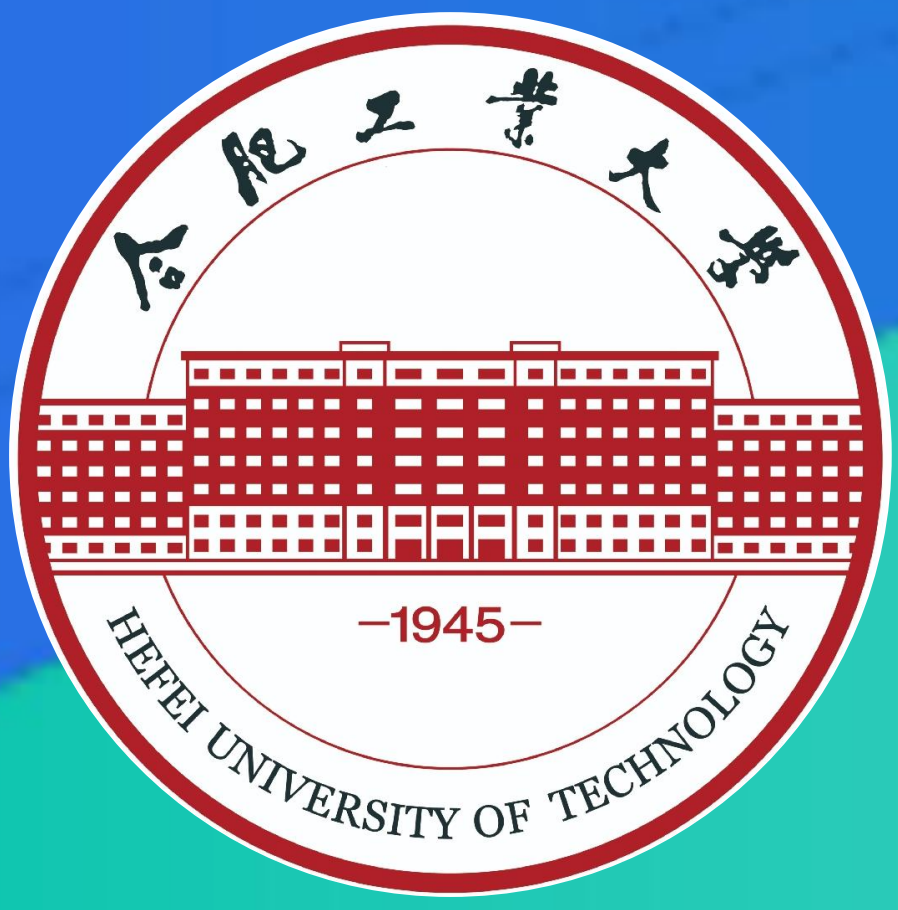




Maskable Retentive Network for Video Moment Retrieval

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Abstract

Video Moment Retrieval (MR) tasks involve predicting the moment described by a given natural language or spoken language query in an untrimmed video. In this paper, we propose a novel Maskable Retentive Network (MRNet) to address two key challenges in MR tasks: cross-modal guidance and video sequence modeling. Our approach introduces a **new retention mechanism into the multimodal Transformer architecture**, incorporating modality-specific attention modes. Specifically, we employ the Unlimited Attention for language-related attention regions to maximize cross-modal mutual guidance. Then, we introduce the Maskable Retention for video-only attention region to enhance video sequence modeling, that is, recognizing two crucial characteristics of video sequences: 1) bidirectional, decaying, and non-linear temporal associations between video clips, and 2) sparse associations of key information semantically related to the query. We propose a bidirectional decay retention mask to explicitly model temporal-distant context dependencies of video sequences, along with a learnable sparse retention mask to adaptively capture strong associations relevant to the target event. Extensive experiments conducted on **five popular benchmarks ActivityNet Captions, TACoS, Charades-STA, ActivityNet Speech, and QVHighlights for MR tasks** demonstrate the significant improvements achieved by our method over existing approaches.

Code is available at <https://github.com/xian-sh/MRNet>.

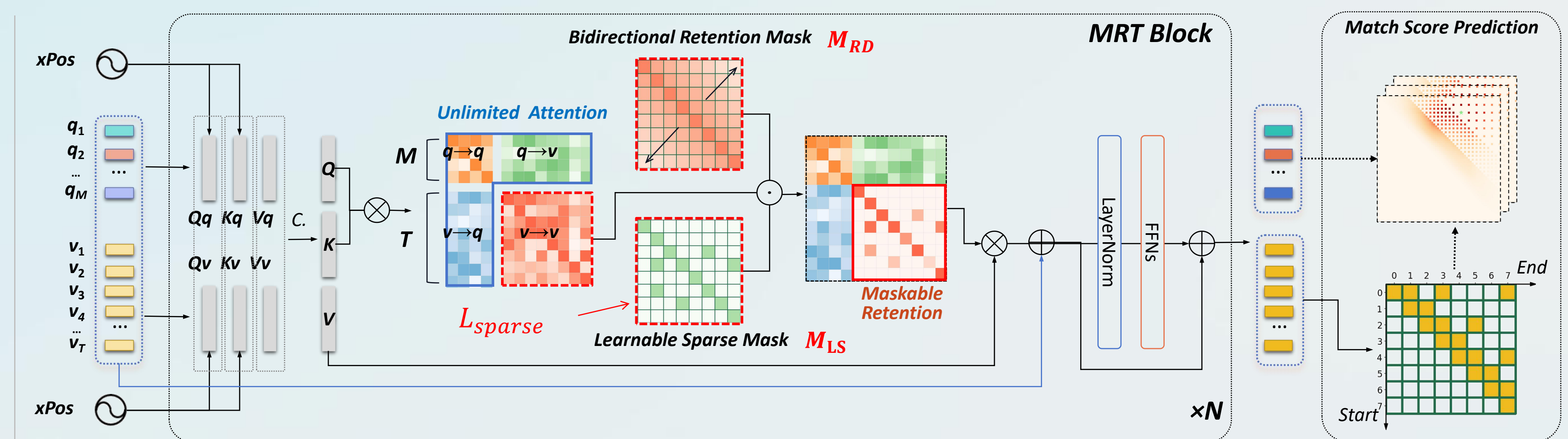
Method: Maskable Retentive Network (MRNet)

NLMR (Text Query): "the man then rides the bike and it moves across the snow."

SLMR (Audio Query):



Ground Truth (GT): 80.75s → 147.34s



An example of query-based video Moment Retrieval: Given a sentence query "the man then rides the bike and it moves across the snow" or an audio query, the goal of MR tasks is to find the moment that is semantically described by the query. We address the visual contextual association learning challenge based on characteristics: ① Temporal association between video clips, ② Redundant background interference.

The representation of language video attention A:

$$A = \begin{bmatrix} Q_q \\ Q_v \end{bmatrix} \cdot [K_q^T K_v^T] = \begin{bmatrix} A_{M \times M} & A_{M \times T} \\ A_{T \times M} & A_{T \times T} \end{bmatrix}$$

learnable sparse mask M_{LS} and spars constraint loss L_{sparse} ,

$$M_{LS} = (1 - I_{T \times T}) \cdot \sigma(W_{LS}) + I_{T \times T}$$

$$L_{sparse} = \frac{1}{T^2} \sum_{i=1}^T \sum_{j=1}^T |\sigma(W_{LS})_{i,j}|$$

A single-layer MRT block:

$$X' = X + \text{MaskRetention}(X)$$

$$\text{MRT}(X) = X' + \text{FFN}(\text{LayerNorm}(X'))$$

Details of the Method: We first introduce a new retention mechanism into the multimodal Transformer architecture. We divide the language and video attention map into four regions: A ($q \rightarrow q$), A ($q \rightarrow v$), A ($v \rightarrow q$) and A ($v \rightarrow v$), so that we can perform modality-specific operations based on the properties of the modality itself: (1) As the linguistic semantics are highly condensed and in order to better maximize the guiding role of language, we set the **Unlimited Attention mode for the language-related regions** A ($q \rightarrow q$), A ($q \rightarrow v$) and A ($v \rightarrow q$), allowing queries can utilize visual details without restriction, and each video clip can response to each query, thus achieving cross-modal mutual guidance; (2) As for the pure video branch A ($v \rightarrow v$), we propose a new **Maskable Retention mechanism to model two characteristics of video sequences together**. In addition, to ensure the sparsity of the video attention, we design a self-supervised sparsity constraint loss to remove redundant responses.

Experiments

Datasets
 ANetCap
 TACoS
 Charades-STA
 ANetSpeech
 QVHighlights
Evaluation metrics
 Recall
 mIoU
 mean average precision (mAP)

Results on ANetCap and TACoS Datasets

Method	Backbones	Text	ANetCap						TACoS					
			R@1	IoU@		R@5	IoU@	mIoU	R@1	IoU@		R@5	IoU@	mIoU
			0.3	0.5	0.7	0.3	0.5	0.7	0.1	0.3	0.5	0.1	0.3	0.5
VSLNet [63]	Transformer	GloVe	63.16	43.22	26.16	-	-	-	43.19	-	-	29.61	24.27	-
2D-TAN [66]	CNN	GloVe	59.45	44.51	26.54	85.53	77.13	61.96	-	47.59	37.29	25.32	70.31	57.81
CSMGAN [25]	BNN/GCN	GloVe	49.11	29.15	77.43	59.63	-	33.90	27.09	53.98	41.22	-	-	24.11
CPNet [16]	Transformer	GloVe	-	40.56	21.63	-	-	-	40.65	-	42.51	28.29	-	-
VSLNet-L [62]	Transformer	GloVe	62.35	43.86	27.51	-	-	-	44.06	-	47.11	36.34	-	36.61
MS-2D-TAN [65]	CNN	GloVe	61.04	46.16	29.21	87.30	78.80	60.85	-	49.24	41.74	34.29	78.33	67.01
MSAT [64]	Transformer	-	-	48.02	31.78	-	78.02	63.18	-	-	48.79	37.57	-	67.63
RaNet [9]	GCN/CNN	GloVe	-	45.59	28.67	-	75.93	62.97	-	-	43.34	33.54	-	67.33
FVMN [10]	Transformer	GloVe	60.63	45.00	26.85	86.11	77.42	61.04	-	53.12	41.48	29.12	78.12	64.53
MMN [52]	CNN	BERT	65.05	48.59	29.26	87.25	79.50	64.76	-	51.39	39.24	26.17	78.03	62.03
MGPN [44]	CNN/Transformer	GloVe	-	47.92	30.47	-	78.15	63.56	-	-	48.81	36.74	-	71.46
DCLN [61]	CNN	GloVe	65.58	44.41	24.80	84.65	74.04	56.67	-	65.16	44.96	28.72	82.40	66.13
SPL [23]	Transformer/GCN	GloVe	-	52.89	32.04	-	82.65	67.21	-	-	42.73	32.58	-	64.30
VGCL [53]	Transformer/RNN	GloVe	60.57	42.96	25.68	-	-	-	43.34	-	-	-	-	-
MASSRN [22]	Transformer/GCN	GloVe	-	51.97	31.39	-	84.05	68.11	-	-	47.88	37.65	-	66.02
PTRM [67]	CNN	BERT	66.41	50.44	31.18	-	-	-	47.68	-	-	-	-	-
CRaNet [43]	GCN/CNN	GloVe	-	47.27	30.34	-	78.84	63.51	-	-	47.86	37.02	-	70.78
BLRN [40]	Transformer/CNN	BERT	-	48.47	31.15	-	81.37	64.44	-	-	-	-	-	-
PLN [68]	CNN	GloVe	59.65	45.66	29.28	89.66	76.65	63.06	44.12	53.74	43.89	31.12	75.56	65.11
DFM [50]	CNN	GloVe	58.84	45.92	32.18	-	-	-	-	-	40.04	28.57	-	27.35
MRNet-S (Ours)	Retentive Network	GloVe	71.92	54.91	31.58	91.28	85.98	73.73	51.48	70.01	53.74	37.54	88.05	77.76
MRNet-S (Ours)	Retentive Network	BERT	73.40	55.40	32.46	90.82	85.24	72.23	51.80	69.43	54.49	37.84	87.60	77.26
MRNet-M (Ours)	Retentive Network	GloVe	74.74	59.40	39.12	90.91	85.12	73.45	54.86	71.71	55.41	38.54	89.43	77.18
MRNet-M (Ours)	Retentive Network	BERT	77.57	63.30	42.68	91.60	86.25	75.92	57.62	71.98	56.16	41.31	89.33	78.43

Results on ANetSpeech Dataset

Method	R@1, IoU@				R@5, IoU@				mIoU
	0.3	0.5	0.7		0.3	0.5	0.7		
VSLNet	46.75	29.08	16.24	-	-	-	-	-	34.01
VGCL [53]	49.80	30.05	16.63	-	-	-	-	-	35.36
SIL [51]	49.46	30.26	15.22	82.28	63.73	35.48	34.52	-	34.52
VSLNet†	51.02	30.38	17.45	-	-	-	-	-	37.04
MMN†	51.98	35.69	20.77	85.46	75.29	56.87	37.81	-	37.81
MRNet-S (Ours)	66.72	49.80	29.12	90.24	84.05	72.53	48.12	-	48.12
MRNet-M (Ours)	70.41	54.61	34.00	90.82	85.19	73.10	51.31	-	51.31

Results on QVHighlights Dataset

Method	MR				HD			
	R@1, IoU@	mAP	Avg.	>= Very Good	R@1, IoU@	mAP	HIT@1	
	@0.5	@0.7	@0.5	@0.75	@0.5	@0.7	@0.5	@0.75
MCN [1]	11.41	2.72	24.94	8.22	10.67	-	-	-
CAL [7]	25.49	11.54	23.40	7.65	9.89	-	-	-
CLIP [35]	16.88	5.19	18.11	7.00	7.67	31.30	61.04	-
XML [15]	41.83	30.35	44.63	31.73	32.14	34.49	55.25	-
XML+ [14]	46.69	33.46	47.89	34.67	34.90	35.38	55.06	-
Moment-DETR [14]	52.89	33.02	54.82	29.40	30.73	35.69	55.60	-
UMT [27]	56.23	41.18	53.83	37.01	36.12	38.18	59.99	-
MH-DETR [55]	60.05	42.48	60.75	38.13	38.38	38.22	60.51	-
QD-DETR [29]	62.40	44.98	62.52	39.88	39.86	38.94	62.40	-
UniVTG [21]	58.86	40.86	57.60	35.59	35.47	38.20	60.96	-
MRNet-S (Ours)	64.85	46.63	65.11	42.06	41.63	39.69	63.55	-

Results on Charades-STA Dataset

Method	R@1, IoU@				R@5, IoU@				mIoU
	0.3	0.5	0.7		0.3	0.5	0.7		
DRN [60]	-	53.09	31.75	-	89.06	60.05	-	-	-
TMLGA [38]	67.53	52.02	33.74	-	-	-	-	-	-
LGI [30]	72.96	59.46	35.48	-	-	-	-	-	51.38
BPNet [54]	65.48	50.75	31.64	-	-	-	-	-	46.34
CPNet [16]	-	60.27	38.74	-	-	-	-	-	52.00
MS-2D-TAN [65]	-	60.08	37.39	-	89.06	59.17	-	-	-
FVMN [10]	-	55.01	33.74	-	89.17	57.24	-	-	-
I ² N [31]	-	56.61	34.14	-	81.48	55.19	-	-	-
PLN [68]	68.60	56.02	35.16	94.54	87.63	62.34	49.09	-	-
M ² DCapsN [26]	-	55.03	31.61	-	84.33	63.71	-	-	-
MRNet-S (Ours)	74.17	60.03	38.25	96.29	88.79	69.92	53.01	-	-
MRNet-M (Ours)	74.65	60.30	38.20	96.51	89.09	69.52	53.27	-	-

The effectiveness of MRNet framework has been demonstrated on five standard benchmarks for three MR tasks.

Ablation studies

Ablation of two masks M_{RD} and M_{LS}

M_{RD}	M_{LS}	ANetCap			Charades-STA		
		0.5	0.7	mIoU	0.5	0.7	mIoU
×	×	58.59	38.93	54.72	57.80	36.72	51.25
✓	×	61.06	40.28	56.04	58.95	37.31	52.57
×	✓	61.33	40.39	56.23	59.03	37.28	52.54
✓	✓	63.30	42.68	57.62	60.27	38.39	53.14

Ablation of two masks L_{sparse}

M_{LS}	L_{sparse}	ANetCap			Charades-STA		
		0.5	0.7	mIoU	0.5	0.7	mIoU
×	×	61.06	40.28	56.04	58.95	37.31	52.57
✓	×	62.79	41.86	57.04	60.00	38.23	52.98
✓	✓	63.30	42.68	57.62	60.27	38.39	53.14

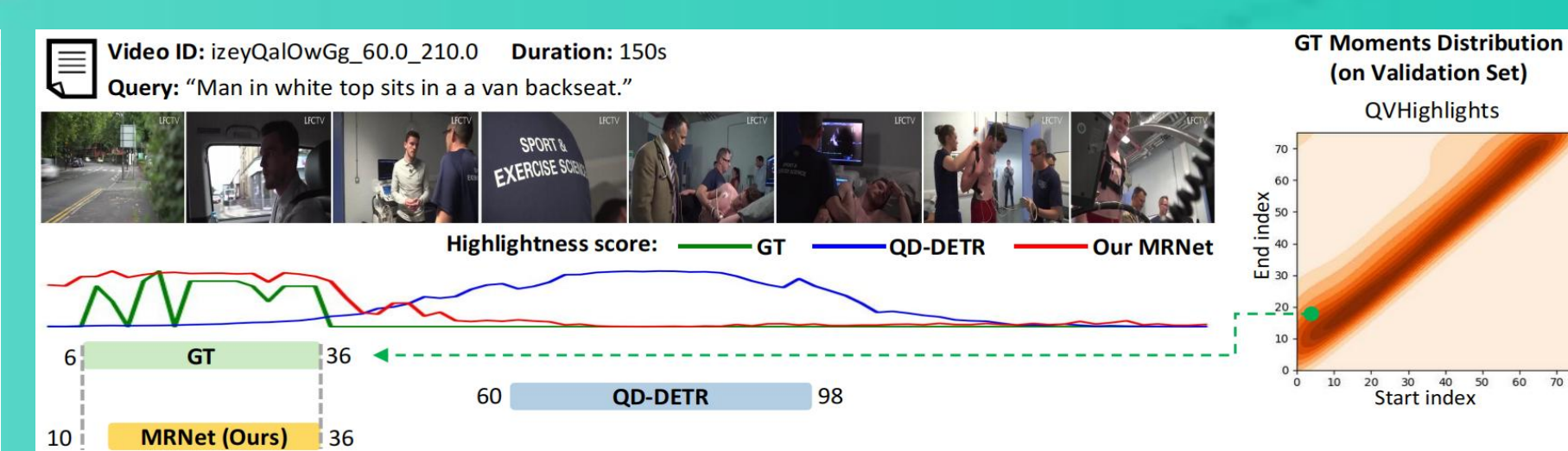
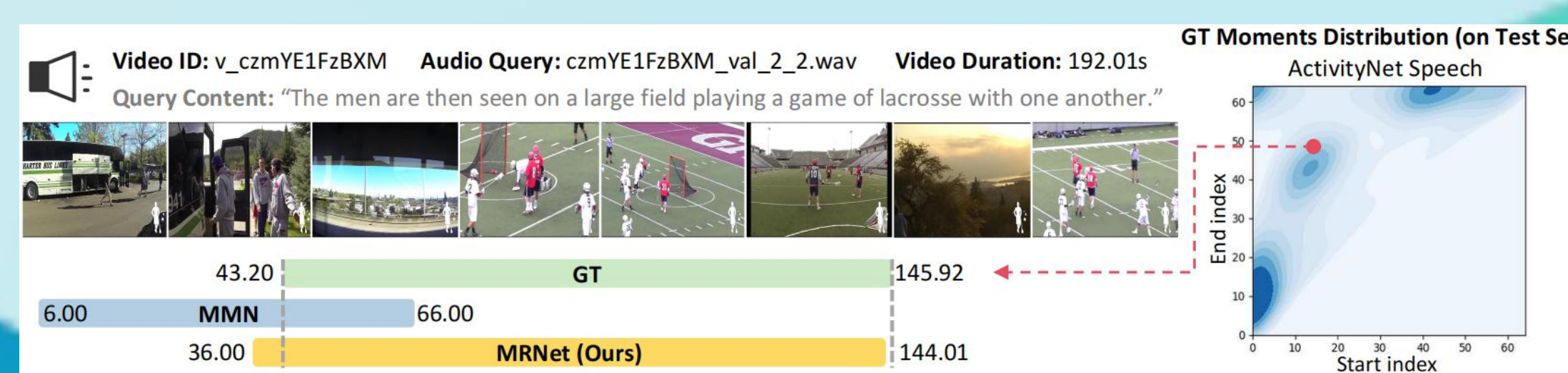
Ablation of masking different regions

$\mathcal{A}(v \rightarrow v)$	$\mathcal{A}(q \rightarrow q)$	$\mathcal{A}(v \rightarrow q)$	$\mathcal{A}(q \rightarrow v)$	IoU0.3	IoU0.5	IoU0.7
×	×	×	×	73.39	58.59	38.93
✓	×	×	×	77.57	63.30	42.68
✓	✓	×	×	76.73	62.18	41.69
✓	✓	✓	×	75.56	61.47	41.02
✓	✓	✓	✓	75.09	60.40	40.39

Convergence curves



Qualitative Results



We take three videos from ANetCap, Charades-STA, ANetSpeech, and QVHighlights datasets as examples and display the qualitative results of them. Compared with the strong MR methods MMN and QD-DETR, our MRNet has almost no gap compared to GroundTruth (GT) across datasets, which indicates that broad adaptability of our approach to MR-related tasks. Overall, the visualization results demonstrate the effectiveness of our proposed MRNet.

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